



REAL ESTATE PROJECT

Regression Analysis

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Arabic Section

Introduction:

The real estate market is a dynamic and significant sector in any economy, influenced by various factors such as location, accessibility, infrastructure, and property characteristics. Understanding the determinants of house prices is essential for buyers, sellers, investors, and policymakers to make informed decisions. This study focuses on analyzing a dataset of real estate properties to identify the key factors that impact the price per unit area of houses.

The dataset consists of data related to real estate properties, with the following variables:¹

- Transaction Date (X_1): The date of the property transaction in a decimal format (e.g., 2012.917 corresponds to a specific month and year).
- House Age (X_2): The age of the house in years.
- Distance to the Nearest MRT Station (X_3): Distance in meters to the nearest Mass Rapid Transit station.
- Number of Convenience Stores (X_4): The count of convenience stores within walking distance from the property.
- Latitude (X_5): Geographic latitude of the property.
- Longitude (X_6): Geographic longitude of the property.
- House Price of Unit Area (Y): The dependent variable representing the price per unit area of the house (in some currency unit).

Objective:

The objective of this study is to build a multiple regression analysis model to explore the relationships between the independent variables (house age, distance to MRT station, number of convenience stores, latitude, and longitude) and the dependent variable (house price of unit area). The transaction date was excluded from the analysis, as it is more suitable for time series analysis rather than regression modeling in this context. This model aims to identify key factors influencing house prices and provide insights for stakeholders in the real estate market.

By achieving this objective, the study aspires to provide actionable insights and evidence-based guidance to real estate stakeholders, aiding in market evaluations and investment decisions.

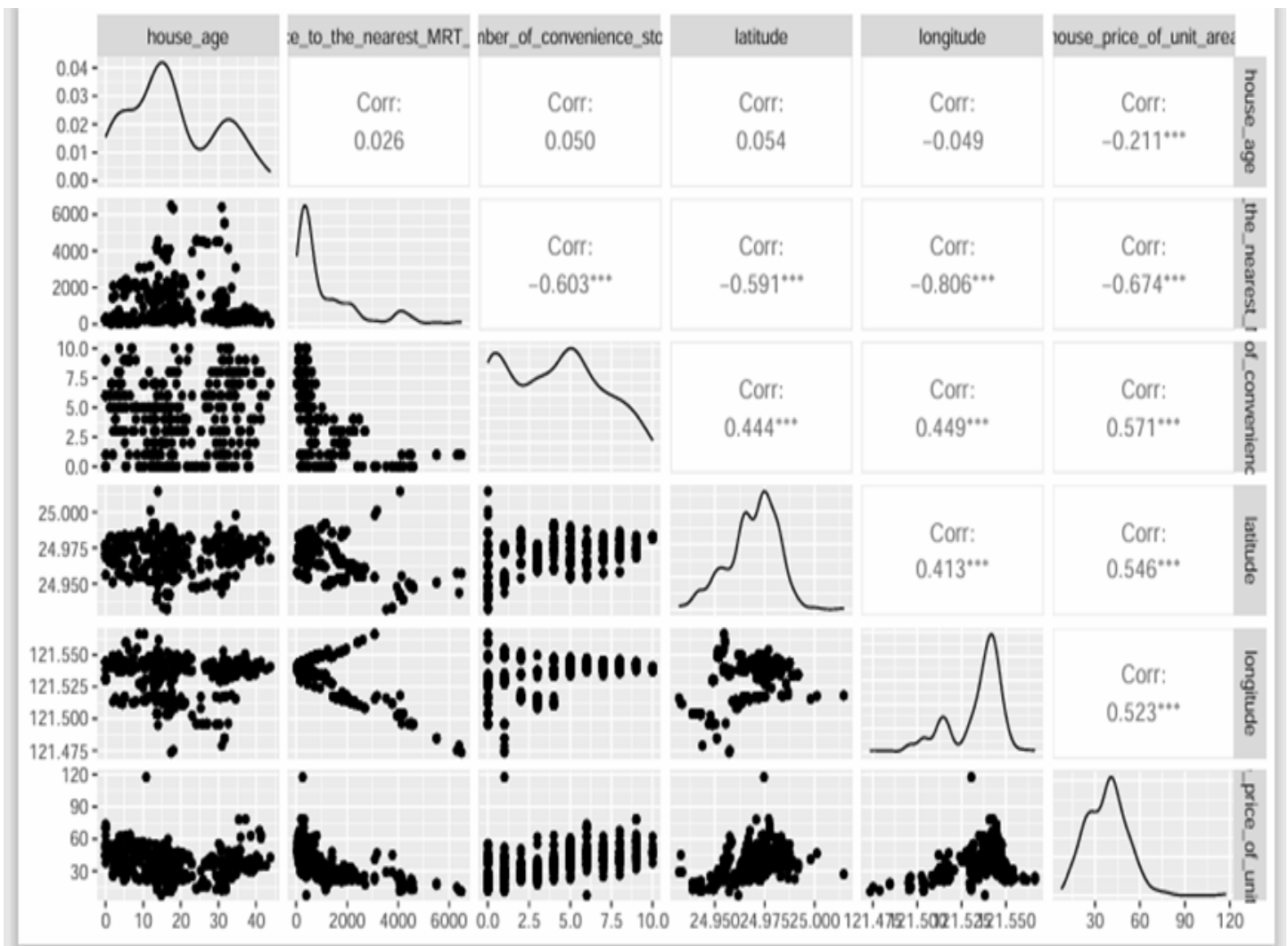
The initial model:

Deterministic assumptions:

- 1- **Parameters are linear in the deterministic part of the model:** Since we are using the linear regression model, this ensures the linearity of the parameters.
- 2- **The values of the explanatory variables are recorded without errors:** As the data has already been collected, we will assume that this assumption holds.

¹ Link of the data: <https://drive.google.com/drive/folders/1e10JglVh93826jfdsYSadUSVQnKpDkxL>

- 3- **The explanatory variables are fixed in repeated samples:** This assumption cannot be verified directly from the data, so we will assume it holds.
- 4- **Reasonable variation in the values of the explanatory variables:** The data for the five explanatory variables has been checked, and none of the explanatory variables takes a constant value.
- 5- **The sample size must be greater than the number of parameters to be estimated:** The data contains 414 data points, and the number of parameters to be estimated is 5 (including the intercept). Therefore, the degrees of freedom are $n - P = 414 - 5 = 409$, which is sufficient for estimating the model.
- 6- **No multicollinearity between the explanatory variable in the multiple regression model:** A correlation matrix was conducted to examine the relationships between six variables: house age, distance to the nearest MRT station, the number of convenience stores, latitude, longitude, and house price of unit area. The correlation coefficients range from -1 and 1, where positive values indicate direct relationships, negative values indicate inverse relationships, and values near zero suggest no linear relationship.



- **House Age and House Price of Unit Area:** A weak negative correlation was observed, $r = -0.211$, $p_value < 0.05$ (statistically significant), suggesting older houses tend to have slightly lower prices.

- **Distance to the Nearest MRT Station and House Price of Unit Area:** A strong negative correlation was found, $r = -0.674$, $p_value < 0.05$ (statistically significant), indicating that houses closer to MRT stations are associated with higher prices.
- **Number of Convenience Stores and House Price of Unit Area:** A moderate positive correlation was noted, $r = 0.571$, $p_value < 0.05$ (statistically significant), suggesting that properties with more convenience stores nearby tend to have higher prices.
- **Latitude and House Price of Unit Area:** A moderate positive correlation was identified, $r = 0.546$, $p_value < 0.05$ (statistically significant) indicating that houses located further north are more expensive.
- **Longitude and House Price of Unit Area:** A moderate positive correlation was observed, $r = 0.523$, $p_value < 0.05$ (statistically significant) suggesting that houses located further east also tend to be more expensive.
- **Distance to the Nearest MRT Station and Number of Convenience Stores:** A very strong negative correlation was observed, $r = -0.603$, $p_value < 0.05$ (statistically significant), indicating that areas closer to MRT stations tend to have a higher number of convenience stores.
- **Latitude and Distance to the Nearest MRT Station:** A moderate negative correlation, $r = -0.591$, $p_value < 0.05$ (statistically significant), suggests that houses further north tend to be closer to MRT stations.
- **House Age and Distance to the Nearest MRT Station:** No significant relationship was observed between house age and distance to the nearest MRT station, $r = 0.026$, $p_value > 0.05$ (statistically insignificant).
- **House Age and Longitude:** No significant relationship was observed between house age and Longitude ($r = -0.049$, $P_value > 0.05$) (statistically insignificant).
- **House Age and Latitude:** No significant relationship was observed between house age and Longitude ($r = 0.054$, $P_value > 0.05$) (statistically insignificant).
- **House Age and Distance to the nearest MRT:** No significant relationship was observed between house age and Distance to the nearest MRT ($r = 0.026$, $P_value > 0.05$) (statistically insignificant).
- **Distance to the nearest MRT and Longitude:** A very strong negative correlation was observed, $r = -0.806$, $p_value < 0.05$ (statistically significant) Houses located further east are generally closer to MRT stations, (general indicator on multicollinearity between Distance to the Nearest MRT Station and Longitude since $r > 0.7$, it is possible that longitude indirectly influences house prices through its association with MRT accessibility).
- **The Number of convenience stores and Longitude:** A moderate positive correlation was observed, ($r = 0.449$, $p_value < 0.05$) (statistically significant) indicating that Houses located further east tend to have more convenience stores nearby.
- **The Number of convenience stores and Latitude:** A moderate positive correlation ($r = 0.444$, $p_value < 0.05$) (statistically significant), indicating that Houses located further north tend to have more convenience stores nearby.
- **Longitude and Latitude:** A moderate positive correlation ($r = 0.413$, $p_value < 0.05$) (statistically significant), indicating that as location of the houses move eastward, they also tend to move slightly northward.

Vif interpretation:

```
> vif(model)

house_age distance_to_the_nearest_MRT_station number_of_convenience_stores
1.014249          4.282985              1.613339

latitude          longitude
1.599017          2.923881
```

The Variance Inflation Factor (VIF) is a measure used to assess multicollinearity among predictors in a regression model. A high VIF indicates that a predictor is highly correlated with other predictors, which can inflate the standard errors of the regression coefficients and reduce the reliability of the estimates.

All predictors in the model have acceptable VIF values ($VIF < 5$), indicating that multicollinearity is not a significant issue, and the coefficients can be interpreted with confidence. The predictor with the highest VIF is longitude ($VIF=2.92$), suggesting possible multicollinearity, likely due to its spatial relationship with latitude or distance to MRT stations.

Model fitting:

A multiple linear regression analysis was conducted to examine the effects of house age, distance to the nearest MRT station, the number of convenience stores, latitude, and longitude on house prices (measured as price per unit area). The model's performance and significance were evaluated using the regression coefficients, R square and p-values.

```
> anova(model)
Analysis of Variance Table

Response: house_price_of_unit_area

            Df Sum Sq Mean Sq  F value    Pr(>F)    
house_age    1   3390    3390   42.1840 2.420e-10 ***
distance_to_the_nearest_MRT_station 1  34164    34164 425.0967 < 2.2e-16 ***
number_of_convenience_stores        1   3817     3817  47.4910 2.106e-11 ***
latitude     1   2299     2299  28.6106 1.478e-07 ***
longitude    1      2        2   0.0252  0.8739    
Residuals   408  32790        80                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Coefficients:

            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -4.946e+03  6.211e+03  -0.796    0.426    
house_age    -2.689e-01  3.900e-02  -6.896 2.04e-11 ***
distance_to_the_nearest_MRT_station -4.259e-03  7.233e-04  -5.888 8.17e-09 ***
number_of_convenience_stores        1.163e+00  1.902e-01   6.114 2.27e-09 ***
latitude     2.378e+02  4.495e+01   5.290 2.00e-07 ***
longitude    -7.805e+00  4.915e+01  -0.159    0.874    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 8.965 on 408 degrees of freedom
Multiple R-squared:  0.5712,    Adjusted R-squared:  0.5659 
F-statistic: 108.7 on 5 and 408 DF,  p-value: < 2.2e-16
```

The regression model was statistically significant, $F(5,408) = 108.7$, $p_value < 0.05$, indicating that the predictors collectively explained a significant amount of variance in house prices. The $R\ square = 0.571$ and **adjusted R square = 0.5669** indicates that **57.1% of the variations in house prices is accounted for by the predictors.**

Coefficients:

Intercept ($\widehat{\beta}_0 = -4946.03$): which indicates that the House Price of Unit area = -4946.03 when all predictors are zero.

House Age ($\widehat{\beta}_1 = -0.2689$): For each additional year of house age, house price decreases by 0.269 units on average, holding all other variables constant.

Distance to the Nearest MRT Station ($\widehat{\beta}_2 = -0.0043$): For every unit increase in distance (meters), house price decreases by 0.0043 units, on average, holding all other variables constant.

Number of Convenience Stores ($\widehat{\beta}_3 = 1.163$): For every additional convenience store, house price increases by 1.16 units on average, holding all other variables constant.

Latitude ($\widehat{\beta}_4 = 237.8$): Houses located further north tend to have higher price.

Longitude ($\widehat{\beta}_5 = -7.805$): Higher longitude values are associated with lower house prices.

Testing the significance of Individual Predictors:

$$H_0: \beta_0 = 0 \quad Vs \quad H_1: \beta_0 \neq 0$$

$$P_value = 0.426 > 0.05$$

Decision: The Null hypothesis will not be rejected at significance level = 5%

The intercept is statistically insignificant, indicating no meaningful baseline house price when all predictors are zero.

$$H_0: \beta_1 = 0 \quad Vs \quad H_1: \beta_1 \neq 0$$

$$P_value = 2.04e-11 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

House Age is statistically significant ($\widehat{\beta}_1 = -0.2689$), indicating that older houses are associated with lower prices per unit area.

$$H_0: \beta_2 = 0 \quad Vs \quad H_1: \beta_2 \neq 0$$

$$P_value = 8.17e-09 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

Distance to the Nearest MRT Station is statistically significant ($\widehat{\beta}_2 = -0.004259$), indicating houses located farther from MRT stations tend to have lower prices per unit area.

$$H_0: \beta_3 = 0$$

Vs

$$H_1: \beta_3 \neq 0$$

$$P_value = 2.27e-9 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

The number of convenience stores ($\widehat{\beta}_3=1.163$) is a significant predictor. The positive coefficient suggests that properties located in areas with more convenience stores tend to have higher prices per unit area.

$$H_0: \beta_4 = 0$$

Vs

$$H_1: \beta_4 \neq 0$$

$$P_value = 2.00e-07 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

The latitude ($\widehat{\beta}_4=237.8$) is statistically significant, suggesting that properties located further north are associated with higher prices per unit area.

$$H_0: \beta_5 = 0$$

Vs

$$H_1: \beta_5 \neq 0$$

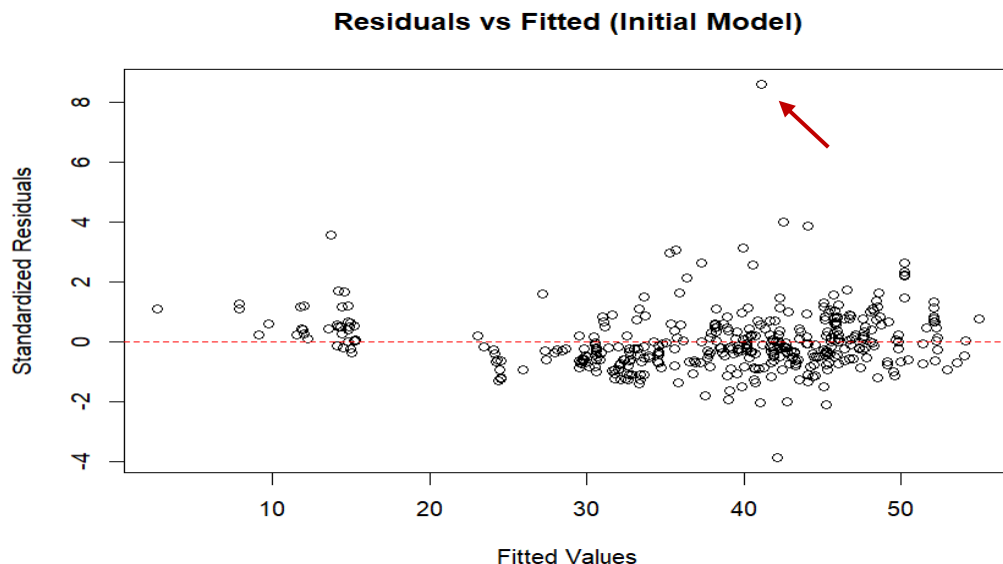
$$P_value = 0.8739 > 0.05$$

Decision: The Null hypothesis will be not rejected at significance level = 5%.

Longitude ($\widehat{\beta}_5 = -7.805$) is not statistically significant, indicating that east-west positioning does not have a meaningful impact on Houses price.

Random assumptions:

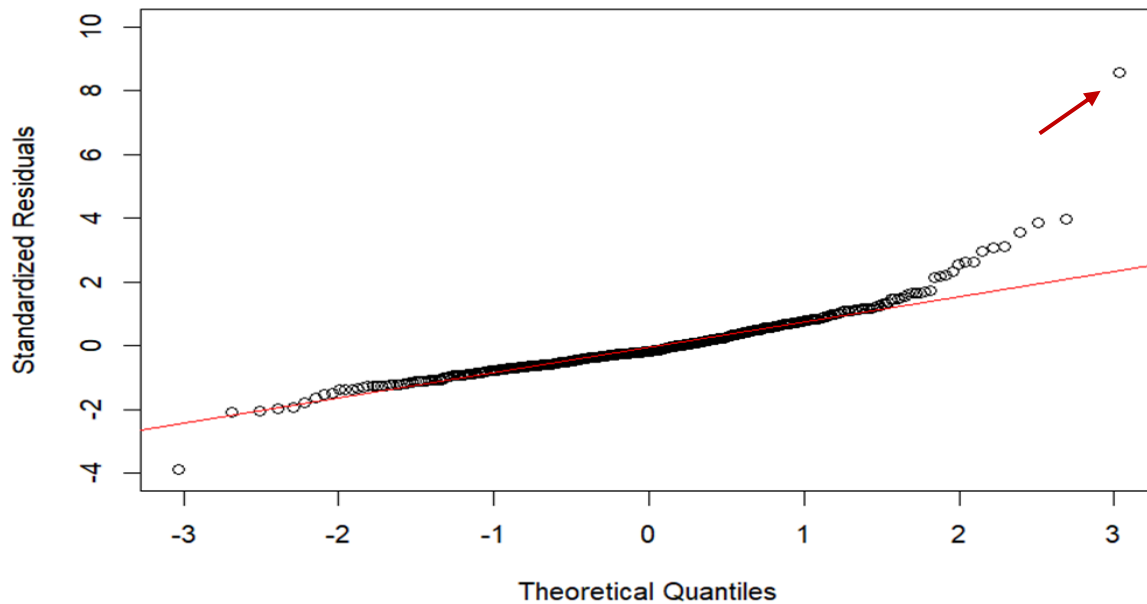
- 1- **The errors are independent, $cov(\epsilon_i, \epsilon_j) = 0$:** In this dataset, there is no intuitive natural order that we know about in the explanatory variables. So, we can assume independence between the errors.
- 2- **The expected value of the random error is zero and the variability in the random error is constant (homoscedasticity):**



The dots are centered around zero but are more concentrated at larger values. However, there are points that are potentially outliers, such as the point indicated by the red arrow. Additionally, there is a funnel shape, which suggests that the variance of errors is not constant. Therefore, we can conclude that the assumptions are not valid.

3- **The errors are normally distributed, $\epsilon_i \sim N(0, \delta^2)$ for all $i = 1, \dots, n$** : we can check this assumption from the Q-Q plot as follows:

Q-Q Plot for Initial Model

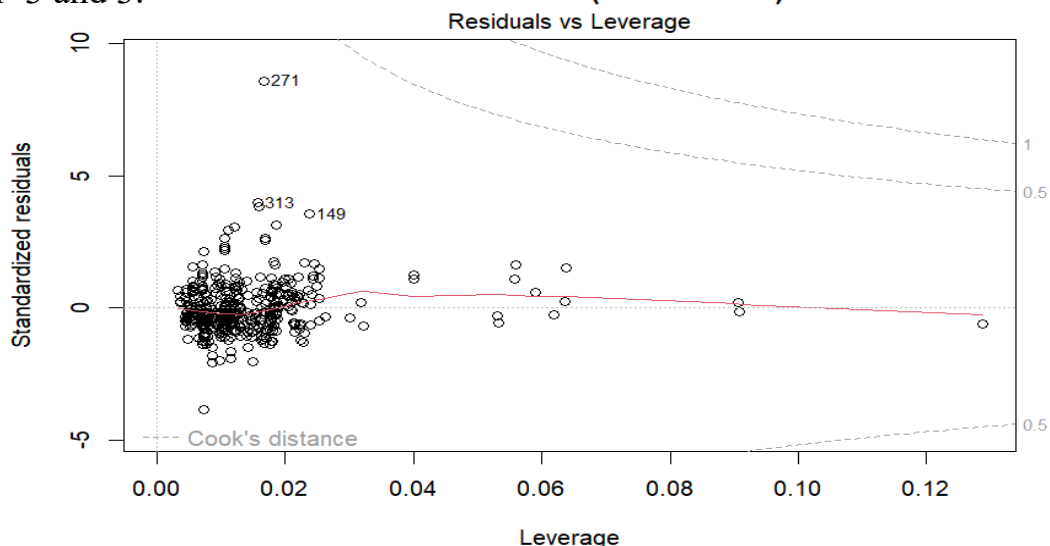


Residuals:

Min	1Q	Median	3Q	Max
-34.546	-5.267	-1.600	4.247	76.372

The figure reveals significant deviations from the Q-Q line, especially in the upper-right section. Additionally, a point marked by a red arrow stands out as a potential extreme outlier. This suggests that the normality assumption does not hold, and the residual range is not between -3 and 3.

Cook's Distance (Initial Model)



AS the plot shows there is no influential points but there is an outliers like point 271.

The reduced Model (Model without longitude):

The removal of the longitude variable did not impact any of the deterministic assumptions, ensuring that all assumptions remain valid. However, we must check for multicollinearity.

```
> vif(model_updated)
             house_age distance_to_the_nearest_MRT_station
             1.013216                               1.992371
number_of_convenience_stores latitude
             1.607857                               1.575344
```

All predictors in the model have acceptable VIF values ($VIF < 5$), indicating that multicollinearity is not a significant issue, and the coefficients can be interpreted with confidence.

Model Fitting:

```
> anova(model_updated)
Analysis of Variance Table

Response: house_price_of_unit_area
              Df Sum Sq Mean Sq F value    Pr(>F)
house_age      1   3390    3390  42.285 2.304e-10 ***
distance_to_the_nearest_MRT_station  1  34164   34164 426.112 < 2.2e-16 ***
number_of_convenience_stores      1   3817    3817  47.605 1.993e-11 ***
latitude        1   2299    2299  28.679 1.428e-07 ***
Residuals    409  32792      80
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> summary(model_updated)

Call:
lm(formula = house_price_of_unit_area ~ house_age + distance_to_the_nearest_MRT_station +
    number_of_convenience_stores + latitude, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-34.522  -5.292  -1.579   4.264  76.466

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -5.916e+03  1.113e+03  -5.317 1.74e-07 ***
house_age     -2.687e-01  3.893e-02  -6.903 1.95e-11 ***
distance_to_the_nearest_MRT_station -4.175e-03  4.928e-04  -8.473 4.37e-16 ***
number_of_convenience_stores    1.165e+00  1.897e-01   6.141 1.94e-09 ***
latitude      2.386e+02  4.456e+01   5.355 1.43e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.954 on 409 degrees of freedom
Multiple R-squared:  0.5711,    Adjusted R-squared:  0.5669
F-statistic: 136.2 on 4 and 409 DF,  p-value: < 2.2e-16
```

The overall regression model statistically significant, $F(4, 409) = 136.2$, $p < 0.05$, indicating that the predictors collectively explain a significant proportion of the variance in the dependent variable. The model had a **Multiple R²** of 0.571, The advariations in567, meaning it explained 56.7 % of the variations in house price per unit area is accounted for by the predictors , Excluding **longitude** improved the model's adjusted R square, F-statistic, and residual standard error, demonstrating that longitude was not a meaningful predictor in the context of this analysis. Removing it enhanced the model's simplicity and statistical efficiency without reducing explanatory power.

Coefficients:

Intercept ($\widehat{\beta}_0 = -5916$): which indicates that the House Price of Unit area = -5916 when all predictors are zero.

House Age ($\widehat{\beta}_1 = -0.269$): For each additional year of house age, house price decreases by 0.269 units on average, holding all other variables constant.

Distance to the Nearest MRT Station ($\widehat{\beta}_2 = -0.0042$): For every unit increase in distance (meters), house price decreases by 0.0042 units, on average, holding all other variables constant.

Number of Convenience Stores ($\widehat{\beta}_3 = 1.165$): For every additional convenience store, house price increases by 1.165 units on average, holding all other variables constant.

Latitude ($\widehat{\beta}_4 = 238.6$): Houses located further north tend to have higher price.

Testing the significance of Individual Predictors:

$$H_0: \beta_0 = 0 \quad Vs \quad H_1: \beta_0 \neq 0$$

$$P_value = 1.74e-07 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%

The intercept is statistically significant, indicating meaningful baseline house price when all predictors are zero.

$$H_0: \beta_1 = 0 \quad Vs \quad H_1: \beta_1 \neq 0$$

$$P_value = 1.95e-11 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

House Age is statistically significant ($\widehat{\beta}_1 = -0.269$), indicating that older houses are associated with lower prices per unit area.

$$H_0: \beta_2 = 0 \quad Vs \quad H_1: \beta_2 \neq 0$$

$$P_value = 4.73e-16 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

Distance to the Nearest MRT Station is statistically significant ($\widehat{\beta}_2 = -0.0042$), indicating properties located farther from MRT stations tend to have lower prices per unit area.

$$H_0: \beta_3 = 0 \quad Vs \quad H_1: \beta_3 \neq 0$$

$$P_value = 1.94e-09 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

The number of convenience stores ($\widehat{\beta}_3 = 1.165$) is a significant predictor. The positive coefficient suggests that properties located in areas with more convenience stores tend to have higher prices per unit area.

$$H_0: \beta_4 = 0 \quad Vs \quad H_1: \beta_4 \neq 0$$

$$P_value = 1.43e-07 < 0.05$$

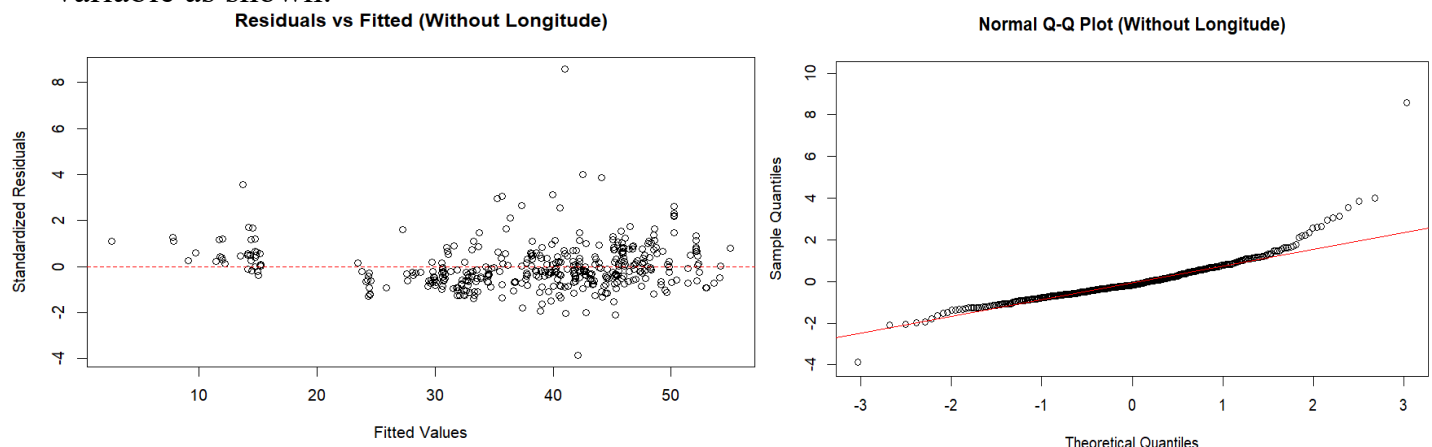
Decision: The Null hypothesis will be rejected at significance level = 5%.

The latitude is statistically significant ($\widehat{\beta}_4 = 238.6$), suggesting that properties located further north are associated with higher prices per unit area.

Comparing the result with the model with longitude there is no significant impact on coefficients values.

Random Assumptions:

1. **The expected value of the random error is zero and the variability in the random error is constant (homoscedasticity):** The results did not change after removing the longitude variable as shown.

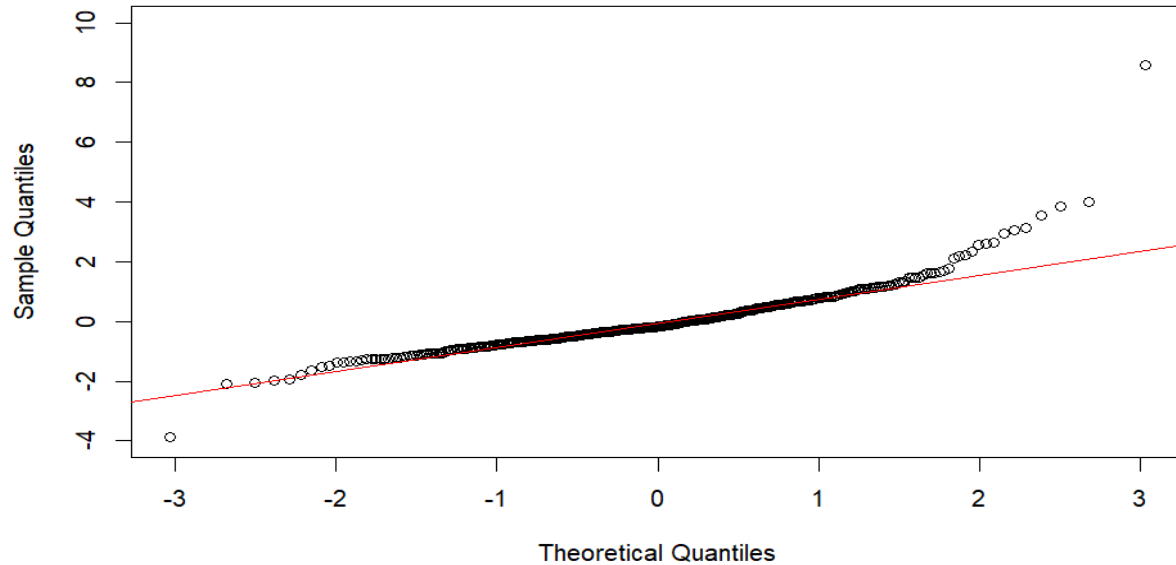


Residuals:

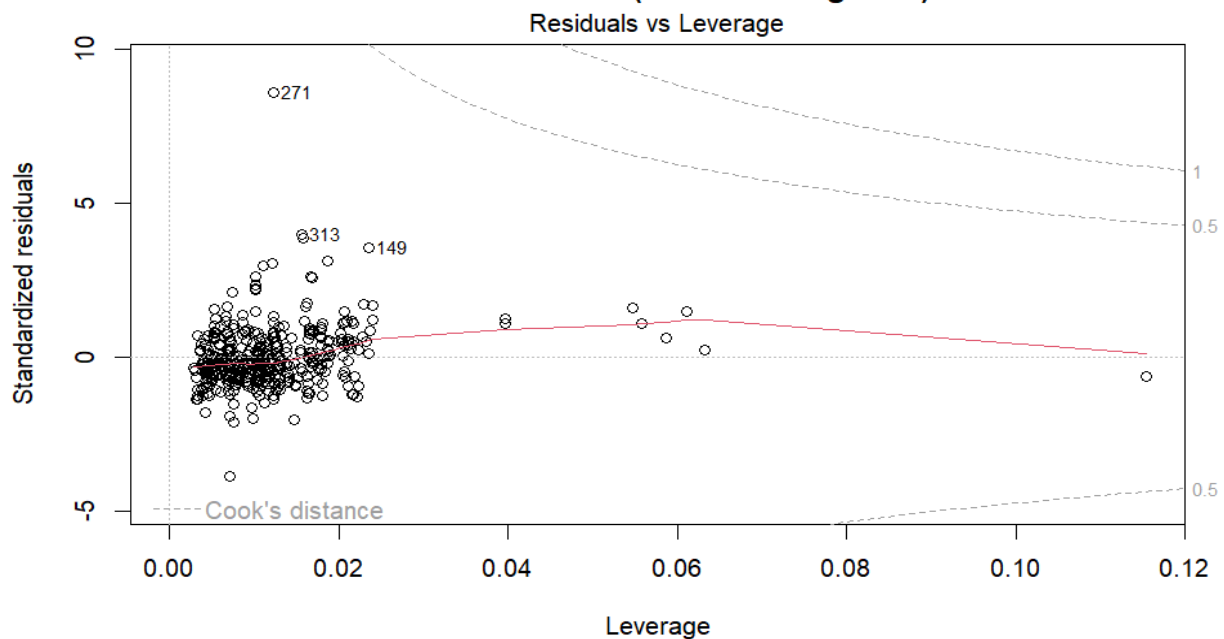
Min	1Q	Median	3Q	Max
-34.522	-5.292	-1.579	4.264	76.466

2. The errors are normally distributed, $\epsilon_i \sim N(0, \delta^2)$ for all $i = 1, \dots, n$:

Normal Q-Q Plot (Without Longitude)



Cook's Distance (Without Longitude)



As shown in the Q-Q plot, the normality remains unchanged from the initial model. Additionally, the Cook's Distance plot indicates the presence of uninfluential outliers, such as point 271.

3. The errors are independent, $cov(\epsilon_i, \epsilon_j) = 0$: the assumption is still hold as removing the longitude variable don't affect it.

The log model:

Taking the log of dependent variable (House Price of Unit Area) did not affect any of the deterministic assumptions, ensuring that all assumptions remain valid. However, we must check for multicollinearity.

```
> vif(log_model)
```

	house_age	distance_to_the_nearest_MRT_station
	1.013216	1.992371
number_of_convenience_stores		latitude
	1.607857	1.575344

All predictors in the model have acceptable VIF values ($VIF < 5$), indicating that multicollinearity is not a significant issue, and the coefficients can be interpreted with confidence.

Fitting Model:

```
> anova(log_model)
```

Analysis of Variance Table

Response: log_house_price

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
house_age	1	2.279	2.279	45.254	5.853e-11	***
distance_to_the_nearest_MRT_station	1	35.568	35.568	706.414	< 2.2e-16	***
number_of_convenience_stores	1	2.465	2.465	48.963	1.073e-11	***
latitude	1	2.711	2.711	53.840	1.181e-12	***
Residuals	409	20.593	0.050			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> summary(log_model)
```

Call:
lm(formula = log_house_price ~ house_age + distance_to_the_nearest_MRT_station +
number_of_convenience_stores + latitude, data = data)

Residuals:

	Min	1Q	Median	3Q	Max
	-1.65305	-0.11773	-0.00352	0.10779	1.07360

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-2.009e+02	2.789e+01	-7.203	2.85e-12	***
house_age	-6.959e-03	9.756e-04	-7.133	4.48e-12	***
distance_to_the_nearest_MRT_station	-1.448e-04	1.235e-05	-11.723	< 2e-16	***
number_of_convenience_stores	2.842e-02	4.753e-03	5.979	4.89e-09	***
latitude	8.194e+00	1.117e+00	7.338	1.18e-12	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2244 on 409 degrees of freedom
Multiple R-squared: 0.6763, Adjusted R-squared: 0.6731
F-statistic: 213.6 on 4 and 409 DF, p-value: < 2.2e-16

A multiple linear regression analysis was conducted after taking the log transformation (log linear model) to examine the effects of house age, distance to the nearest MRT station, the number of convenience stores, latitude on house prices.

$\log(\widehat{\text{House Price}}) = -200.9 + (-0.006959 \times \text{House Age}) + (-0.0001448 \times \text{Distance to Nearest MRT Station}) + (0.02824 \times \text{Number of Convenience Stores}) + (8.194 \times \text{Latitude})$.

The overall regression model was statistically significant, $F(4, 409) = 213.6$, $p < 0.05$, indicating that the predictors collectively explain a significant proportion of the variance in the dependent variable. The model had a **Multiple R^2** of 0.676, The adjusted $R^2 = 0.673$, meaning it explained 67.3 % of the variations in house price per unit area.

Coefficients:

Intercept ($\widehat{\beta}_0 = -200.90$): which indicates that the House Price of Unit area approximately = 0, when all predictors are zero.

House Age ($\widehat{\beta}_1 = -0.00696$): For each additional year of house age, house price decreases by 0.696 percent on average, holding all other variables constant.

Distance to the Nearest MRT Station ($\widehat{\beta}_2 = -0.0001448$): For every unit increase in distance (meters), house price decreases by 0.01448 percent on average, holding all other variables constant.

Number of Convenience Stores ($\widehat{\beta}_3 = 0.02842$): For every additional convenience store, house price increases by 2.842 percent on average, holding all other variables constant.

Latitude ($\widehat{\beta}_4 = 8.194$): Houses located further north tend to have higher prices.

Testing the significance of Individual Predictors:

$$H_0: \beta_0 = 0 \quad Vs \quad H_1: \beta_0 \neq 0$$

$$P_value = 2.85e-12 > 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%

The intercept is statistically significant, indicating meaningful baseline house price when all predictors are zero.

$$H_0: \beta_1 = 0 \quad Vs \quad H_1: \beta_1 \neq 0$$

$$P_value = 4.48e-12 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

House Age is statistically significant ($\widehat{\beta}_1 = -0.00696$), indicating that older houses are associated with lower prices per unit area.

$$H_0: \beta_2 = 0 \quad Vs \quad H_1: \beta_2 \neq 0$$

$$\mathbf{P_value = 2e-16 < 0.05}$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

Distance to the Nearest MRT Station is statistically significant ($\widehat{\beta}_2 = -0.000144$), indicating properties located farther from MRT stations tend to have lower prices per unit area.

$$H_0: \beta_3 = 0 \quad Vs \quad H_1: \beta_3 \neq 0$$

$$\mathbf{P_value = 4.89e-09 < 0.05}$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

The number of convenience stores ($\widehat{\beta}_3=0.02842$) is a significant predictor. The positive coefficient suggests that properties located in areas with more convenience stores tend to have higher prices per unit area.

$$H_0: \beta_4 = 0 \quad Vs \quad H_1: \beta_4 \neq 0$$

$$\mathbf{P_value = 1.18e-12 < 0.05}$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

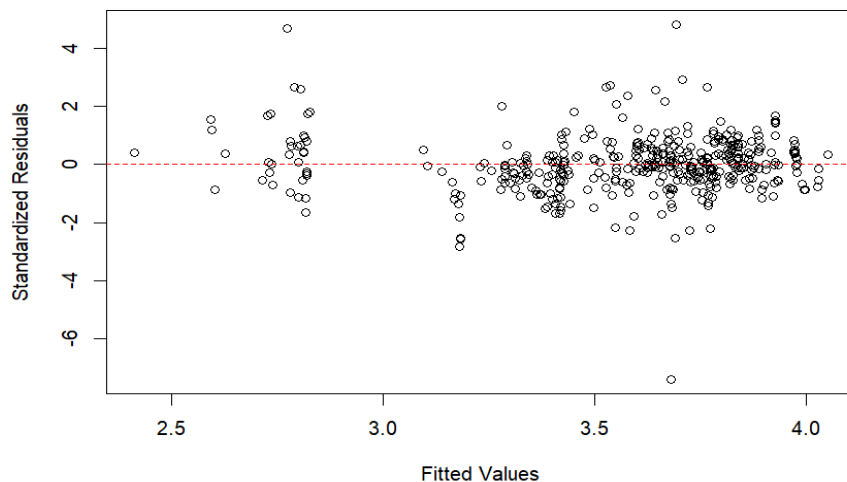
The latitude ($\widehat{\beta}_4=8.194$) is statistically significant, suggesting that properties located further north are associated with higher prices per unit area.

Random Assumptions:

1. The expected value of the random error is zero, $E(\epsilon_i) = 0$ for all $i = 1, \dots, n$ and the variability in the random error is constant (homoscedasticity) $Var(\epsilon_i) =$

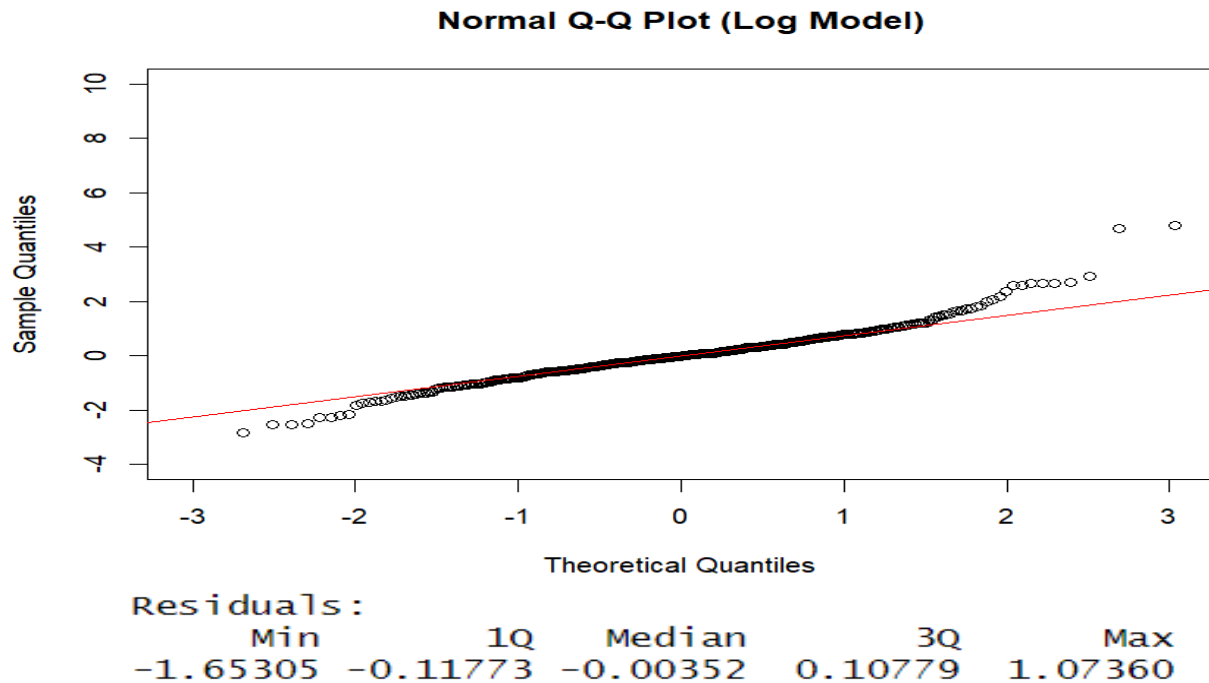
δ^2 for all $i = 1, \dots, n$:

Residuals vs Fitted (Log Model)

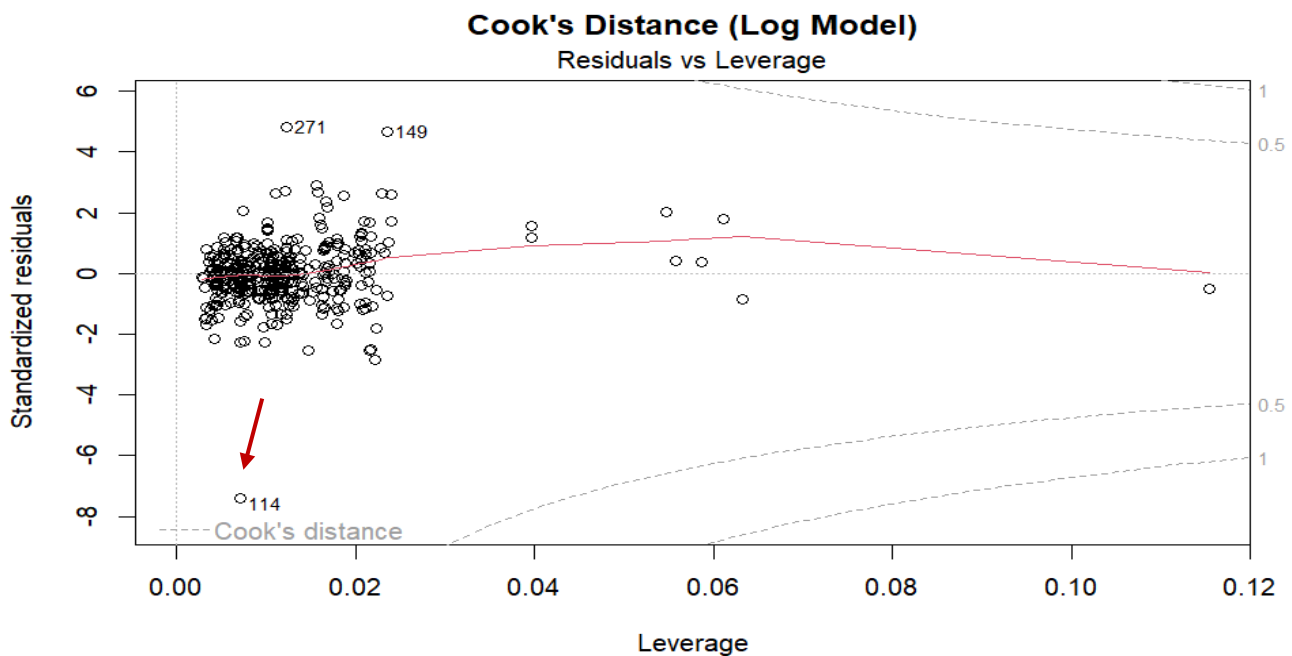


The figure shows that the points are concentrated around zero and remain clustered at higher values due to the log transformation. It is evident that the scale of the fitted values has changed; however, the plot still displays a funnel shape, indicating that the assumptions are not fully met.

2. **The errors are independent, $cov(\epsilon_i, \epsilon_j) = 0$:** the assumption is still hold as taking the log of the dependent variable don't affect it.
3. **The errors are normally distributed, $\epsilon_i \sim N(0, \delta^2)$ for all $i = 1, \dots, n$:**



The normal Q-Q plot shows no significant improvement, with substantial deviations from the line, particularly in the top-right section, and smaller deviations in the bottom-left section. This suggests the presence of outliers that need to be addressed. However, the residuals now fall within the range of -3 to 3, resembling a normal distribution.



The plot shows that point 271 has moved closer to the rest of the data points but remains an insignificant outlier. Additionally, a new point, indicated by the red arrow, has emerged with a strong deviation from the rest of the points.

The log model after removing outliers (the cleaned log model):

Taking the log of dependent variable (House Price of Unit Area) and removing outliers did not affect any of the deterministic assumptions, ensuring that all assumptions remain valid. However, we must check for multicollinearity.

```
> vif(clean_log_model)
             house_age distance_to_the_nearest_MRT_station
             1.013892                      2.003702
number_of_convenience_stores latitude
             1.564551                      1.599035
```

All predictors in the model have acceptable VIF values ($VIF < 5$), indicating that multicollinearity is not a significant issue, and the coefficients can be interpreted with confidence.

Model Fitting:

A multiple linear regression was conducted to predict after eliminating some outlier's observations but not influential ones to predict **house price per unit area** based on four predictors: **house age**, **distance to the nearest MRT station**, **number of convenience stores**, and **latitude**.

```
> anova(clean_log_model)
Analysis of Variance Table

Response: log_house_price

              Df Sum Sq Mean Sq  F value    Pr(>F)
house_age      1  2.431    2.431   101.529 < 2.2e-16 ***
distance_to_the_nearest_MRT_station  1 33.714   33.714  1407.761 < 2.2e-16 ***
number_of_convenience_stores        1  1.652    1.652    69.001 1.743e-15 ***
latitude       1  1.749    1.749    73.032 3.115e-16 ***
Residuals    381  9.124    0.024
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> summary(clean_log_model)

Call:
lm(formula = log_house_price ~ house_age + distance_to_the_nearest_MRT_station +
    number_of_convenience_stores + latitude, data = clean_data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.49157 -0.09743  0.00049  0.09717  0.48243

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.788e+02  2.137e+01  -8.368 1.13e-15 ***
house_age    -8.260e-03  7.117e-04 -11.605 < 2e-16 ***
distance_to_the_nearest_MRT_station -1.768e-04  9.777e-06 -18.084 < 2e-16 ***
number_of_convenience_stores      2.505e-02  3.409e-03   7.347 1.25e-12 ***
latitude      7.313e+00  8.558e-01   8.546 3.12e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1548 on 381 degrees of freedom
Multiple R-squared:  0.8125,    Adjusted R-squared:  0.8106
F-statistic: 412.8 on 4 and 381 DF,  p-value: < 2.2e-16
```

$\log(\widehat{\text{House Price}}) = -178.8 + (-0.00826 \times \text{House Age}) + (-0.0001768 \times \text{Distance to Nearest MRT Station}) + (0.002505 \times \text{Number of Convenience Stores}) + (7.313 \times \text{Latitude})$.

The overall regression model was statistically significant, $F(4, 381) = 412.8, p < 0.05$, indicating that the predictors collectively explain a significant proportion of the variance in the dependent variable. The model had a **Multiple R^2** of 0.812, The adjusted $R^2 = 0.811$, meaning it explained 81.1 % of the variations in house price per unit area.

Coefficients:

Intercept ($\widehat{\beta}_0 = -178.8$): which indicates that the House Price of Unit area approximately = 0, when all predictors are zero.

House Age ($\widehat{\beta}_1 = -0.00826$): For each additional year of house age, house price decreases by 0.826 percent on average, holding all other variables constant.

Distance to the Nearest MRT Station ($\widehat{\beta}_2 = -0.0001768$): For every unit increase in distance (meters), house price decreases by 0.0178 percent on average, holding all other variables constant.

Number of Convenience Stores ($\widehat{\beta}_3 = 0.002505$): For every additional convenience store, house price increases by 0.2505 percent on average, holding all other variables constant.

Latitude ($\widehat{\beta}_4 = 7.313$): Houses located further north tend to have higher prices.

Testing the significance of Individual Predictors:

$$H_0: \beta_0 = 0 \quad Vs \quad H_1: \beta_0 \neq 0$$

$$P_value = 1.13e-15 > 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%

The intercept is statistically significant, indicating meaningful baseline house price when all predictors are zero.

$$H_0: \beta_1 = 0 \quad Vs \quad H_1: \beta_1 \neq 0$$

$$P_value = 2e-16 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

House Age is statistically significant ($\widehat{\beta}_1 = -0.00826$), indicating that older houses are associated with lower prices per unit area.

$$H_0: \beta_2 = 0 \quad Vs \quad H_1: \beta_2 \neq 0$$

$$P_value = 2e-16 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

Distance to the Nearest MRT Station is statistically significant ($\widehat{\beta}_2 = -0.0001768$), indicating properties located farther from MRT stations tend to have lower prices per unit area.

$$H_0: \beta_3 = 0 \quad Vs \quad H_1: \beta_3 \neq 0$$

$$P_value = 1.25e-12 < 0.05$$

Decision: The Null hypothesis will be rejected at significance level = 5%.

The number of convenience stores ($\widehat{\beta}_3 = 0.002505$) is a significant predictor. The positive coefficient suggests that properties located in areas with more convenience stores tend to have higher prices per unit area.

$$H_0: \beta_4 = 0 \quad Vs \quad H_1: \beta_4 \neq 0$$

$$P_value = 3.12e-16 < 0.05$$

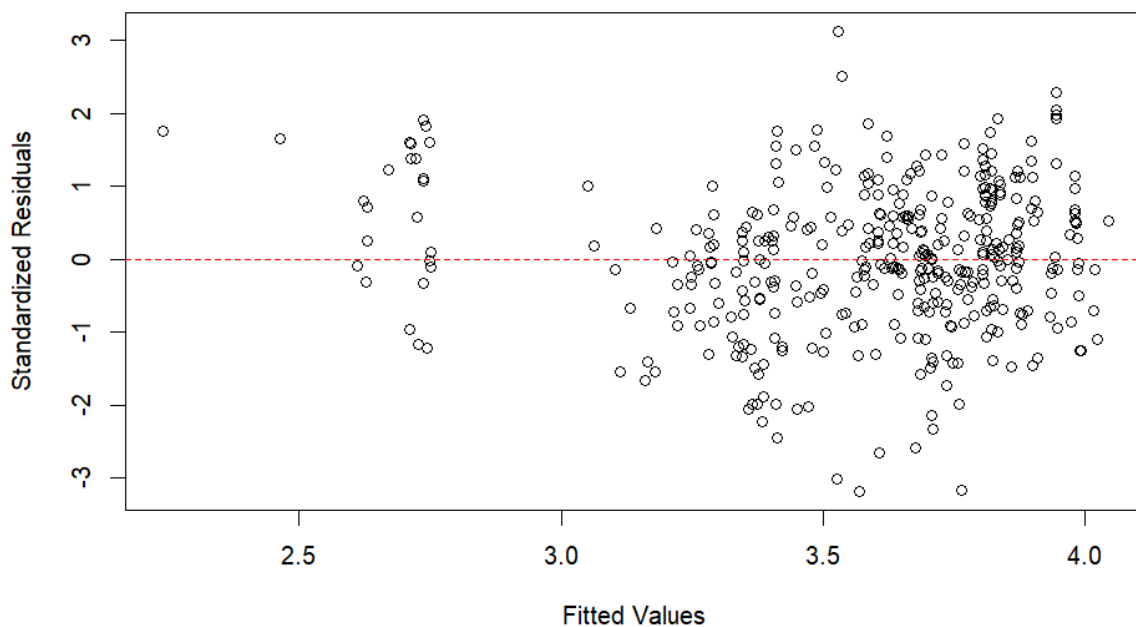
Decision: The Null hypothesis will be rejected at significance level = 5%.

The latitude ($\widehat{\beta}_4 = 7.313$) is statistically significant, suggesting that properties located further north are associated with higher prices per unit area.

Random Assumptions:

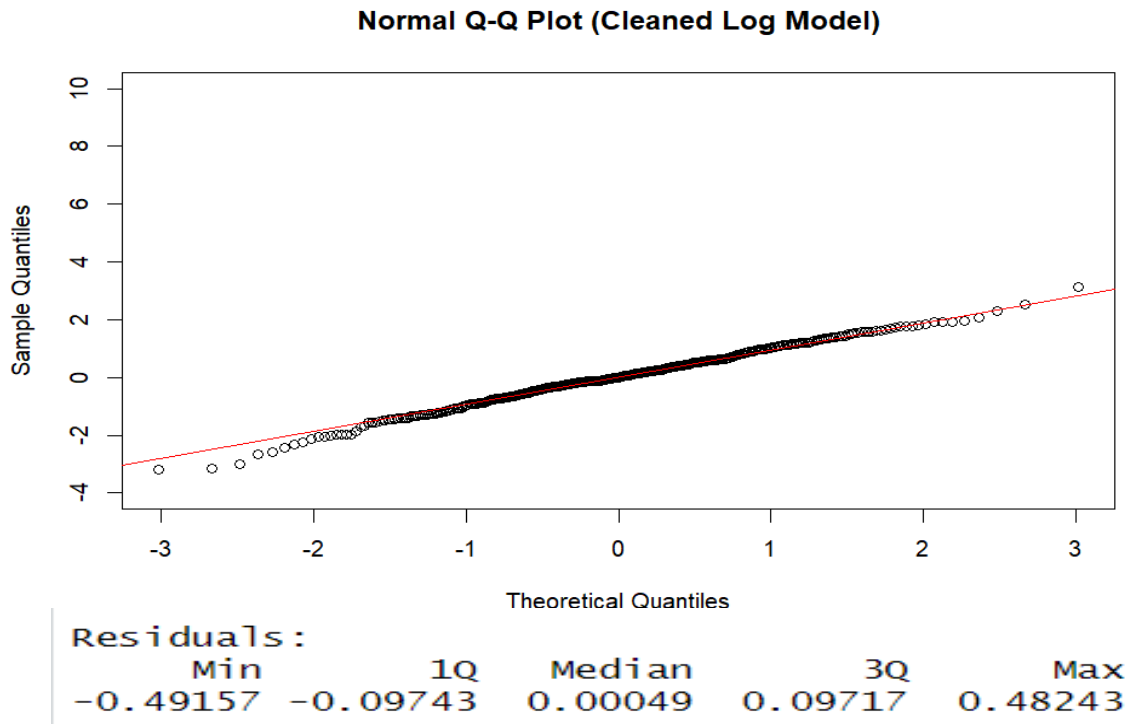
1. The expected value of the random error is zero, $E(\epsilon_i) = 0$ for all $i = 1, \dots, n$ and the variability in the random error is constant (homoscedasticity) $Var(\epsilon_i) = \delta^2$ for all $i = 1, \dots, n$:

Residuals vs Fitted (Cleaned Log Model)

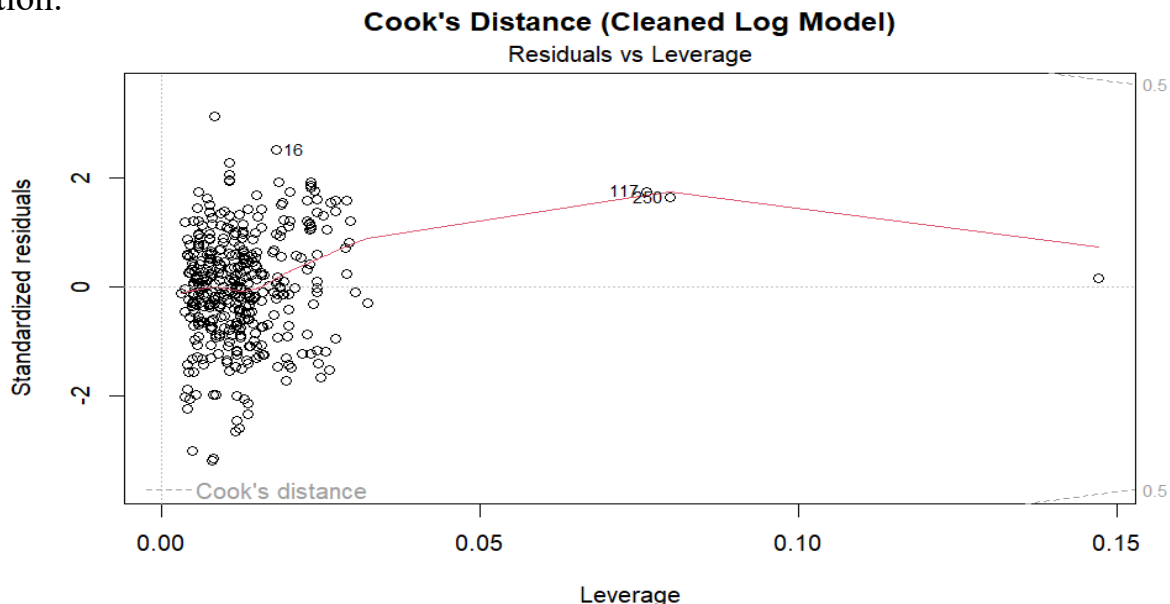


The points are now randomly scattered around the zero line, showing improvement after removing insignificant outliers identified by Cook's distance. This suggests that the expectation assumption is valid; however, the slight funnel pattern indicates that the variance assumption is still not met.

2. The errors are normally distributed, $\epsilon_i \sim N(0, \delta^2)$ for all $i = 1, \dots, n$:



The normal Q-Q plot shows improvement, with no deviations in the top-right section and smaller deviations in the bottom-left section. The remaining points align well with the Q-Q line, indicating that the normality assumption is satisfied. Additionally, the residuals now fall between -3 and 3, and the median has become 0.00049, which suggests symmetry, resembling a normal distribution.



The Cook's distance plot shows significant improvement, with the outliers becoming much less pronounced.

3. The errors are independent, $cov(\epsilon_i, \epsilon_j) = 0$: the assumption is still hold as taking the log of the dependent variable and deleting the outliers don't affect it.

Conclusion with Coefficient Interpretation:

The final regression model, after removing outliers, successfully predicts house price per unit area based on four key predictors: house age, distance to the nearest MRT station, number of convenience stores, and latitude. The equation is:

$$\log(\text{house price per unit area}) = -178.8 - 0.00826(\text{House age}) - 0.0001768(\text{Distance to the nearest MRT station}) + 0.002505(\text{number of convenience stores}) + 7.313(\text{latitude})$$

- **Intercept ($\beta_0 = -178.8$):** This represents the baseline log-transformed house price when all predictors are zero. Though not directly interpretable in the original scale, it serves as a reference point for the effects of the predictors.
- **House Age ($\beta_1 = -0.00826$):** For each additional year of house age, house price per unit area decreases by 0.826%, holding all other variables constant. This suggests that older houses tend to be cheaper compared to newer ones.
- **Distance to Nearest MRT Station ($\beta_2 = -0.0001768$):** For each meter increase in distance from the nearest MRT station, house price decreases by 0.0177%, holding all other variables constant. This indicates that properties further away from public transport stations are less expensive, highlighting the importance of location and access to transportation.
- **Number of Convenience Stores ($\beta_3 = 0.002505$):** Every additional convenience store in the area increases the house price per unit area by 0.2505%, holding all other variables constant. The presence of more convenience stores enhances the attractiveness of the location, making it more desirable and thus increasing property prices.
- **Latitude ($\beta_4 = 7.313$):** For each unit increase in latitude, house price increases by 731.3%, holding all other variables constant. This suggests a geographical pattern, where houses further north (higher latitude) tend to have higher prices, possibly reflecting a trend towards more desirable or premium areas.

Model Performance and Assumptions:

The model explains 81.1% of the variance in house price per unit area ($R^2 = 0.811$), indicating a strong fit. The overall F-statistic (412.8, $p < 0.05$) confirms that the predictors collectively provide significant explanatory power for house prices.

- **Multicollinearity:** The VIF values for all predictors were below 5, indicating that multicollinearity is not a concern and the coefficient estimates are reliable.
- **Assumptions:** The model met the key assumptions of normality, independence, and homoscedasticity. A slight funnel pattern remains in the residuals, suggesting some variance issues. However, Cook's distance plot showed improvement after removing outliers, confirming that influential points no longer significantly affect the model.

In conclusion, this cleaned log model, with its statistically significant predictors, provides valuable insights into the factors influencing house prices per unit area. It suggests that both physical (e.g., house age, distance to MRT) and locational (e.g., convenience stores, latitude) factors play important roles in determining property values. This model is a useful tool for real estate professionals and buyers to understand and predict price trends.

Appendix:

```
# Load necessary packages
library(car)
library(GGally)

# Import data
data <- read.csv("D:/Regression/regression project/Real estate.csv")

# Initial data exploration: Correlation matrix
ggpairs(data)

# Step 1: Fit the model without log transformation
model <- lm(house_price_of_unit_area ~ house_age + distance_to_the_nearest_MRT_station +
            number_of_convenience_stores + latitude + longitude, data = data)

# Regression summary and diagnostics for the initial model
summary(model)
anova(model)
vif(model)

# Residuals vs Fitted
plot(rstandard(model) ~ fitted(model), main = "Residuals vs Fitted (Initial Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")

# Q-Q plot
qqnorm(rstandard(model), main = "Normal Q-Q Plot (Initial Model)", ylim = c(-4, 10))
qqline(rstandard(model), col = "red")

# Cook's distance
plot(model, which = 5, main = "Cook's Distance (Initial Model)")

# Step 2: Remove 'longitude' as it is insignificant
model_updated <- lm(house_price_of_unit_area ~ house_age +
                    distance_to_the_nearest_MRT_station +
                    number_of_convenience_stores + latitude, data = data)

# Regression summary after removing 'longitude'
summary(model_updated)
anova(model_updated)
vif(model_updated)

# Residuals vs Fitted
plot(rstandard(model_updated) ~ fitted(model_updated), main = "Residuals vs Fitted (Without
Longitude)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")

# Q-Q plot
qqnorm(rstandard(model_updated), main = "Normal Q-Q Plot (Without Longitude)", ylim = c(-4,
10))
qqline(rstandard(model_updated), col = "red")

# Cook's distance
plot(model_updated, which = 5, main = "Cook's Distance (Without Longitude)")

# Step 3: Log-transform the dependent variable and fit the model
```

```

data$log_house_price <- log(data$house_price_of_unit_area)
log_model <- lm(log_house_price ~ house_age + distance_to_the_nearest_MRT_station +
               number_of_convenience_stores + latitude, data = data)

# Summary and diagnostics for the log-transformed model
summary(log_model)
anova(log_model)
vif(log_model)

# Residuals vs Fitted
plot(rstandard(log_model) ~ fitted(log_model), main = "Residuals vs Fitted (Log Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")

# Q-Q plot
qqnorm(rstandard(log_model), main = "Normal Q-Q Plot (Log Model)", ylim = c(-4, 10))
qqline(rstandard(log_model), col = "red")

# Cook's distance
plot(log_model, which = 5, main = "Cook's Distance (Log Model)")

# Step 4: Identify and remove outliers based on Cook's Distance
cooks_d <- cooks.distance(log_model)
potential_outliers <- which(cooks_d > (4 / nrow(data))) # Rule of thumb threshold

# Removing outliers
clean_data <- data[-potential_outliers, ]

# Step 5: Refit the log-transformed model with cleaned data
clean_log_model <- lm(log_house_price ~ house_age + distance_to_the_nearest_MRT_station +
                    number_of_convenience_stores + latitude, data = clean_data)

# Summary of the cleaned log-transformed model
summary(clean_log_model)
anova(clean_log_model)
vif(clean_log_model)

# Diagnostics for the cleaned log-transformed model
# Residuals vs Fitted
plot(rstandard(clean_log_model) ~ fitted(clean_log_model),
     main = "Residuals vs Fitted (Cleaned Log Model)",
     xlab = "Fitted Values", ylab = "Standardized Residuals")
abline(h = 0, lty = 2, col = "red")

# Q-Q plot for cleaned log-transformed model
qqnorm(rstandard(clean_log_model), main = "Normal Q-Q Plot (Cleaned Log Model)", ylim = c(-4,
10))
qqline(rstandard(clean_log_model), col = "red")

# Cook's distance plot for cleaned model
plot(clean_log_model, which = 5, main = "Cook's Distance (Cleaned Log Model)")

# Correlation matrix for cleaned data
ggpairs(clean_data)

```